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# Time-dependent current-voltage characteristics of Al/p-CdTe/Pt x-ray detectors

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Current-voltage (I-V) characteristics of Schottky Al/p-CdTe/Pt detectors were investigated in dark and at different temperatures. CdTe detectors with Al rectifying contacts, very appealing for high resolution x-ray and gamma ray spectroscopy, suffer from bias-induced polarization phenomena which cause current increasing with the time and severe worsening of the spectroscopic performance. In this work, we studied the time-dependence of the I-V characteristics of the detectors, both in reverse and forward bias, taking into account the polarization effects. The I-V measurements, performed at different time intervals between the application of the bias voltage and the measurement of the current, and the measured current transients show as the electrical instability manifests itself even in time intervals shorter (i.e.,  $<1$  s) than those are necessary to make evident the degradation of the spectroscopic properties. The results point out as this time interval is a critical parameter for correct investigations on the electrical properties of these devices. Although the detectors are typically classified as Schottky-type detectors, we find that the thermionic emission is not the dominant transport mechanism, except at very low bias voltage values. Finally, the Schottky barrier height of the Al/p-CdTe contact was extracted by measuring the contact resistance at zero bias. © 2012 American Institute of Physics. [<http://dx.doi.org/10.1063/1.4764325>]

## I. INTRODUCTION

The appealing potentialities of cadmium telluride (CdTe) and cadmium zinc telluride (CdZnTe) detectors for x-ray and gamma ray spectroscopy are now widely recognized.<sup>1–3</sup> Among the traditional semiconductor spectrometers, CdTe detectors show high detection efficiency and good room temperature performance and are well suited for the development of compact and portable detection systems.<sup>4</sup> Beside their successful astrophysical applications,<sup>1,2</sup> a great effort has recently been devoted to the development of CdTe detectors for medical applications, such as x-ray spectroscopy in mammography<sup>5,6</sup> and energy resolved photon counting detectors in diagnostic medicine (mammography and computed tomography).<sup>7–9</sup> High-resolution CdTe detectors are generally fabricated with rectifying contacts, working as diodes with indium (In) as the anode electrode on a p-type CdTe wafer and platinum (Pt) as the cathode.<sup>1</sup> Rectifying contacts ensure low leakage currents, if compared with an ohmic electrode configuration (Pt/CdTe/Pt), allowing high bias voltage operation and optimum charge collection. Due to the difficulties of dividing an In electrode into pixels, aluminum (Al) has been recently found to be a good alternative electrode material for diode detectors.<sup>10–12</sup> In addition to low leakage currents comparable to those of standard In/CdTe/Pt detectors, Al/CdTe/Pt detectors allow pixellization of anodes and then the possibility to realize electron-collecting-type diode detectors.

As well known, CdTe detectors with rectifying contacts (Al and In) suffer from instability with time under bias

voltage, causing dramatic worsening in the spectroscopic performance.<sup>13–15</sup> Several studies demonstrated as this time instability in CdTe detectors, known as polarization, is mainly related to the accumulation of negative charge on deep acceptor levels during the application of the bias voltage (charge accumulation model, CAM).<sup>13–19</sup> Polarization produces a progressive increasing of the reverse current with the time which causes a severe degradation of the energy resolution of the detectors. Studies on polarization phenomena in Schottky CdTe detectors are typically performed through an electrical approach.<sup>17–21</sup> At the end of 2006, Toyama and co-workers<sup>17</sup> presented some results on the physical properties of deep acceptors and the Schottky barrier height (Al/CdTe) by measuring the current voltage curves of Al/CdTe/Pt detectors and on the basis of the CAM model. In this work, the time dependence of the leakage current is considered only for long times (the order of minutes). In Ref. 17, the extraction of the barrier height of the Al/CdTe contact is obtained from the straight line intercept of a semi-log forward I-V plot at zero bias, supposing that the I-V curve at the starting time ( $t=0$ ) is not affected by the polarization. In Ref. 22, the time instability of the leakage current of Schottky CdTe detectors was characterized by measurements in a 30 s/100 h time scale after the bias application. In Ref. 19, the polarization Schottky CdTe detectors were studied in simulated structures, under dark bias, solving drift-diffusion and Poisson equations numerically. The results of the current transient after biasing for Al/CdTe/Pt and In/CdTe/Pt detectors were obtained with a starting time of the order of a few seconds and shorter times are not taken into account in order to speed up the simulation time.

In this work, we presented electrical investigations on Schottky Al/p-CdTe/Pt detectors pointing out the

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time-dependence of the current-voltage characteristics even at short times (i.e.,  $<1$  s) after the bias application. The time dependence of the I-V curves in both reverse and forward bias and at low temperature was showed. Due to the absence of time-dependence at low bias voltages (near zero Volt), we also estimated the barrier height of the Al/CdTe Schottky from the temperature dependence of the specific contact resistance. Furthermore, from the analysis of the reverse current transient, we obtained some clarifications about the effect of the polarization on the transport mechanism.

## II. EXPERIMENTAL PROCEDURES

The study was carried out on p-type CdTe planar detectors, manufactured by *Acrorad* (Japan) with 2 mm thickness and  $4.1 \times 4.1 \text{ mm}^2$  area. The detectors are characterized by Au/Ti/Al/CdTe/Pt electrode configuration. The anode (Schottky) contact was formed by deposition Au/Ti/Al on the Te-face of the crystal. A guard ring is placed around the anode and  $50 \mu\text{m}$  away from it. The cathode is a planar electrode (Pt) covering the entire detector surface, which produces an nearly ohmic contact on the CdTe.<sup>23</sup> A sample of three detectors was characterized. The detector was enclosed in a shielded box placed on a Peltier thermal stage with temperature control to within  $0.1^\circ\text{C}$ , and filled with nitrogen gas to prevent condensation. The detector was mounted on Polytetrafluoroethylene (PTFE) substrate (by *Taconic*) to prevent leakage currents.

The I-V curves were measured with the Keithley 2410, which provides the cathode bias voltage, and with the Keithley 236 configured as electrometer, which is connected to the pixel anode forced at zero bias and measures the pixel current with accuracy  $0.3\% \text{reading} + 100 \text{ fA}$  with  $\pm 1.0 \text{ nA}$  range. The guard ring of the detector is held at the same zero potential of the anode by connecting it at the HI Output guard of the triax connector of the 236 unit. Therefore, we prevent to measure the leakage currents that could flow through any lateral conductive path of the detector.

The I-V measurements were performed in sweep mode, with the 2410 and 236 units synchronized via external trigger link lines. In particular, when the 2410 unit set its source value send a trigger to the 236 unit, which starts the source-delay-measure cycle. At the end of the A/D conversion, the 236 send the trigger to the 2410 unit to start the setting of the new value. The delay time  $t_{\text{delay}}$  between the setting of the source voltage and the starting of the current measuring phase can be varied from 0 to 65 s. The measuring phase comprises the integration time of the A/D conversion, which was chosen equal to 17 ms, which was a good compromise between speed and noise immunity. The total measurement time obtained by measuring the lag between the two consecutive trigger pulses with  $t_{\text{delay}} = 0$  is  $\approx 30 \text{ ms}$ . Hence, the total time delay  $T_{\text{delay}}$  between the time of the setting source voltage and the time of the end of the A/D conversion is  $T_{\text{delay}} = t_{\text{delay}} + 30 \text{ ms}$ .  $T_{\text{delay}}$  includes the trigger latency ( $200 \mu\text{s}$ ), source configuration time ( $50 \mu\text{s}$ ) and the settling time ( $100 \mu\text{s}$ ) of both SMU units.

In order to measure the transient current of the detectors at fixed bias voltage, within a time less than 30 ms after the voltage bias is settled, we used a transimpedance amplifier,

realized with the AD549 ultra low bias current operational amplifier, with  $1.0 \text{ nA/V}$  gain. The detector voltage bias was supplied with the ORTEC 456 H high voltage power supply. The current detector waveforms were acquired with the NI 4551 16-bit data acquisition board, by using  $10^3 \text{ Sample/s}$ . The input voltage range was varied according to the range of the detector current, which depends on the temperature. The waveform record length is  $\approx 1000 \text{ s}$ . The transient measurements were performed by applying a step reverse voltage at the detector. Before the application of the step voltage, the detector is biased at zero volt for a time long enough to avoid residual polarization effects. The voltage step pulse was filtered by means low-pass RC circuit with  $\approx 50 \mu\text{s}$  rise time.

## III. MEASUREMENTS AND RESULTS

### A. DC characterization

We performed the I-V measurements at various temperatures and for different values of the  $T_{\text{delay}}$  (40, 130, and 1030 ms). In Fig. 1 are shown the I-V curves at different temperatures in the range  $-25$ – $40^\circ\text{C}$  and in the voltage range  $-1000$ – $100 \text{ V}$  measured with a the delay time  $T_{\text{delay}} = 1.0 \text{ s}$ . The rectifying behavior of the detectors is clearly highlighted. The reverse currents are very low, especially if compared to the Pt/CdTe/Pt detector ones, allowing high bias voltage operation of the detectors.<sup>24,25</sup> We note that the I-V curves at forward bias and for all temperatures do not present a linear trend in the semi-log scale. Thus, the forward current  $I_F$  not obey to the relation  $I_F \cong I_0 \exp(qV_F/k_B T \eta)$  valid for Schottky contacts, where  $V_F$  is the forward bias voltage,  $q$  is the magnitude of electronic charge,  $k_B$  is the Boltzmann constant,  $T$  is the absolute temperature,  $I_0$  is the reverse saturation current and  $\eta$  is the ideality factor, which usually assumes value close to 1 for metal-semiconductor Schottky contacts.<sup>26,27</sup> The ideality factor  $\eta$  for the characterized detectors assumes values greater than 2 at  $V_F \approx 1 \text{ V}$  and increases with the increasing of the forward bias voltage.

Fig. 2 shows the forward I-V curves of Fig. 1 on log-log scale. We note that at all temperatures the current tends towards a linear trend at high voltages. Thus, at high forward bias, i.e.,  $V_F > 40 \text{ V}$ , the I-V curves follow a power law,

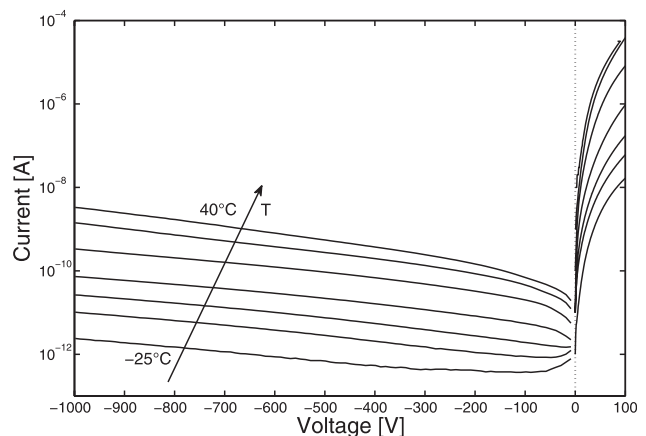


FIG. 1. I-V curves of the Al/p-CdTe/Pt detector measured at  $T = -25, -10, 0, 10, 25, 35$ , and  $40^\circ\text{C}$ , with  $T_{\text{delay}} = 1.0 \text{ s}$ .

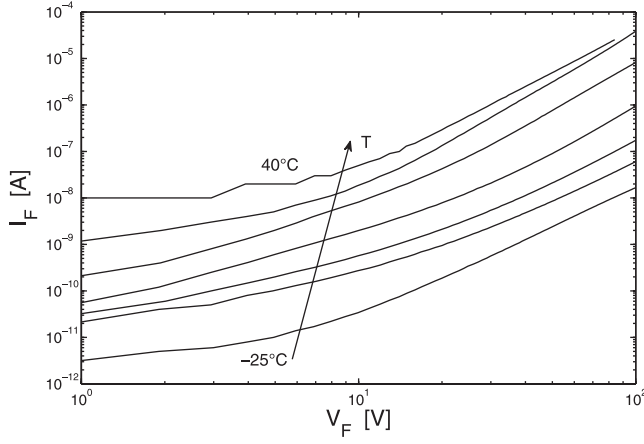


FIG. 2. Forward I-V curves of Fig. 1 on log-log scale. The plateau at  $T = 40^\circ\text{C}$  and at low voltages is due to the high current range of the instrument Keithley 236.

$I_F \propto V_F^m$ , where the exponent  $m$  depends on the bias voltage, the temperature and  $T_{\text{delay}}$ . Generally, if  $m$  assumes values greater than 2 then the conduction mechanism is dominated by space charge limited current (SCLC).<sup>28,29</sup> By changing  $T_{\text{delay}}$  the I-V curves maintain the power-law relation but with different  $m$  values. In Fig. 3 are presented the forward I-V characteristics at different temperatures and for the selected  $T_{\text{delay}}$ . We observe the change of the I-V with the time  $T_{\text{delay}}$ . From the linear fitting in a log-log scale in the high voltage range  $V_F > 40\text{V}$ , we obtain the values of the exponent  $m$ , which are time dependent. For example, at  $T = 25^\circ\text{C}$  we obtain the values  $m = 2.63 \pm 0.03$ ,  $3.05 \pm 0.03$  and  $3.44 \pm 0.02$  for 40, 130 and 1030 ms time delay, respectively. Moreover, the  $m$  versus  $T$  curves change with the acquisition time  $T_{\text{delay}}$ , for example, for  $T_{\text{delay}} = 1.0\text{ s}$ ,  $m$  increases monotonically from 2.8 to 3.6 with the temperature, whereas at  $T_{\text{delay}} < 140\text{ ms}$  the curve  $m(T)$  has a minimum at  $10^\circ\text{C}$ , where  $m \approx 2.5$ .

Also in the reverse bias region the detectors are characterized by time-dependent I-V curves. In Fig. 4 are shown the measured I-V curves for different temperatures and delay times. We note that the reverse current  $I_R$  is very sensitive to the delay time. At low temperatures  $I_R$  decreases with the increasing of  $T_{\text{delay}}$  in the whole voltage bias range. At high temperatures an increasing of the reverse current with the time delay was observed, due to the occurrence of the polarization. The results of this investigation show that the I-V curves measured with delay time greater than  $\approx 1\text{ s}$  and at high reverse bias voltages  $V_R < -200\text{ V}$  follow the law  $I_R(V) = a \exp(b\sqrt[4]{|V_R|})$ , predicted by the thermionic emission model for reversed biased Schottky barriers,<sup>27</sup> although in our case the  $a$  and  $b$  are coefficients depend on both time and the temperature.

## B. Transient characterization

Transient measurements were performed in reverse by measuring the time dependence of the current  $I_R(t)$  within a time window of approximately 1000 s after the detector is switched from zero to a specified reverse bias voltage. Because at reverse bias and at low temperatures the detector current is very small, we performed transient analysis only in the temperature and reverse voltages where the current is greater than resolution of the amplifier AD549 ( $\approx 10\text{ pA}$ ). Currents greater than  $10\text{ nA}$  cause the saturation of the amplifier. In Fig. 5 are shown the transient current at  $V_R = -500\text{ V}$  at different temperatures in the range  $10\text{--}35^\circ\text{C}$ . We note that at all temperatures an initial high current value is present, followed by a current decreasing. The current reaches a minimum value at the time  $t_p$ , which we denote polarization time. For  $t > t_p$  the current  $I_R$  increases with the time. For  $t < 20\text{ ms}$ , the current is a constant due to the amplifier saturation or to the clamping of the analog input stage of the acquisition board.

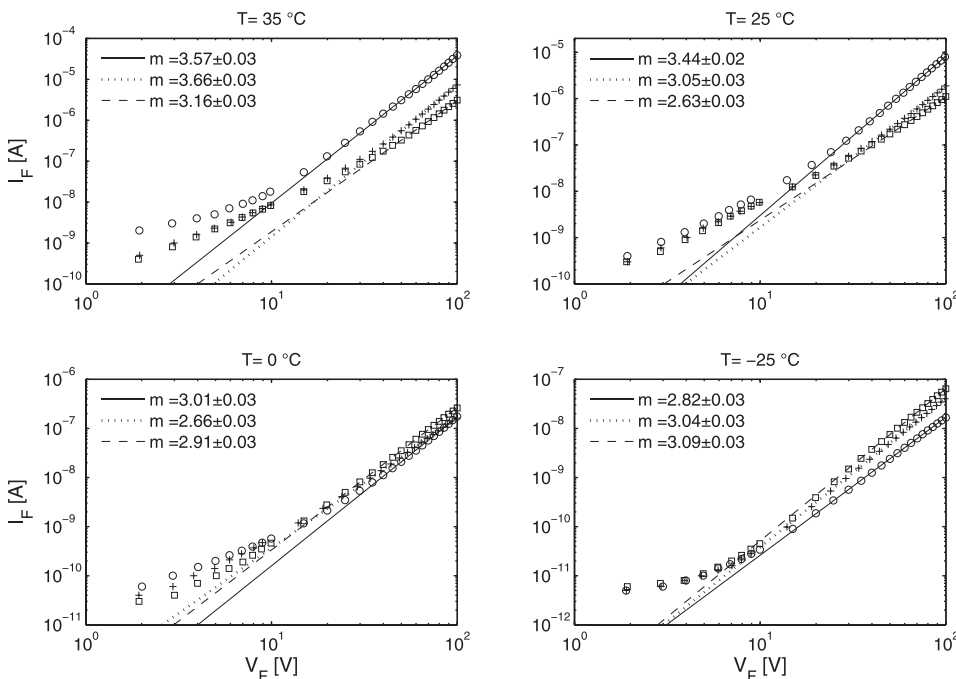


FIG. 3. Time evolution of the forward I-V curves at different temperatures and time  $T_{\text{delay}}$ : (○) 1 s, (+) 130 ms and (□) 40 ms. The straight lines are the fitting obtained with the power-law  $I_F \propto V_F^m$  in the high bias voltage range, i.e.,  $40 < V_F < 100\text{ V}$ .

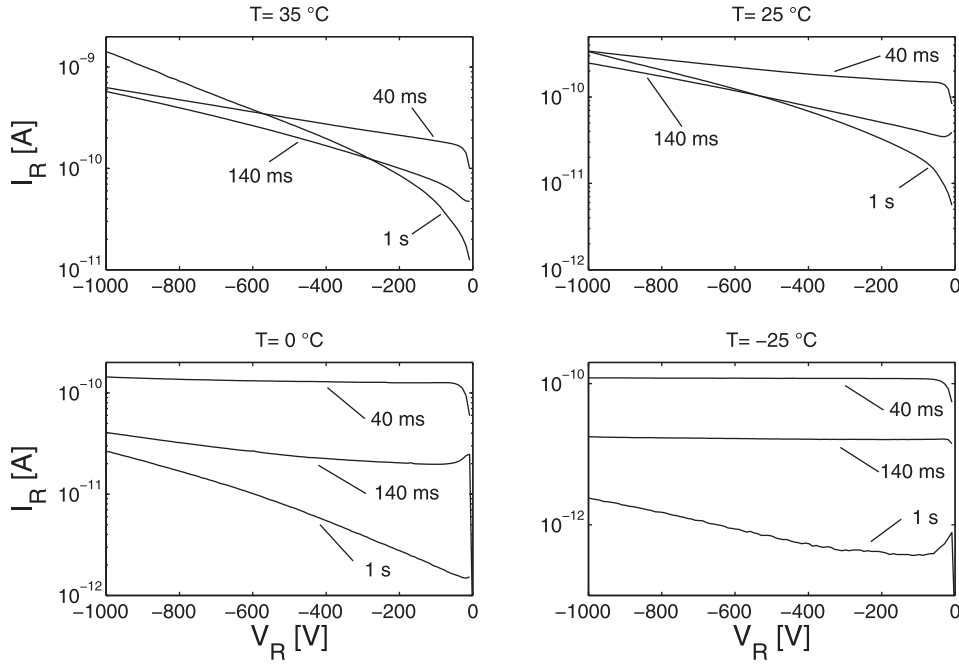


FIG. 4. Time evolution of the reverse I-V curves at different temperatures and delay times  $T_{\text{delay}}$ .

### C. Barrier height

The time dependence of I-V curves occurring even for short ( $<1$  ms) delay times do not allow to extract the barrier height of the Schottky contact Al/CdTe by using the common methods based on the I-V-T analysis.<sup>17,26,27</sup> We noted that at bias voltages less than ( $\approx 1$  V) the I-V curves do not depend on the time  $T_{\text{delay}}$ . Hence, to extract the barrier height  $\phi_{Bp}$  of the hole-blocking Schottky contact Al/CdTe, we use the specific contact resistance  $R_c$  defined as:<sup>27</sup>

$$R_c = \left( \frac{\partial I}{\partial V} \right)_{V=0}^{-1}. \quad (1)$$

By considering that the Pt/CdTe junction is an hole-injecting contact<sup>15,19,30</sup> we neglect its contribute to the contact resistance. If the thermionic emission dominates the current transport, the contact resistance can be expressed as:

$$R_c = \frac{k_B}{qA^*T} \exp\left(\frac{q\phi_{Bp}}{k_B T}\right), \quad (2)$$

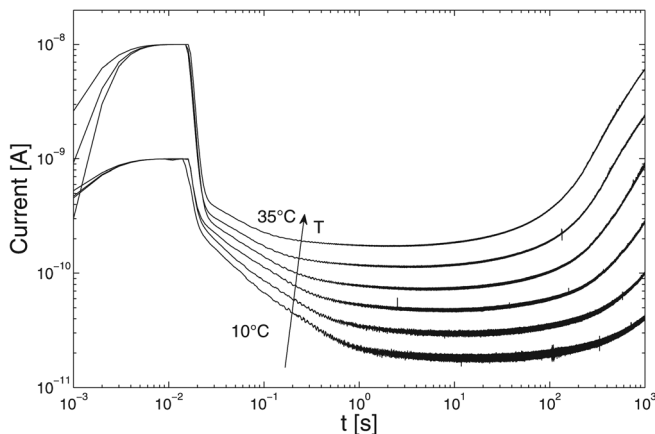


FIG. 5. Transient current curves under reverse bias  $V_R = -500$  V.

where  $A^*$  is the effective Richardson constant for the thermionic emission. We performed current-voltage measurements in the  $-0.1$ – $0.1$  V range with 5 mV step and in the temperature range  $25$ – $60$  °C with  $5$  °C step. The measured I-V curves, excluding the values close to  $V=0$ , are interpolated with a third-degree polynomial, i.e.,  $I = q_1 V^3 + q_2 V^2 + q_3 V + q_4$ . The contact resistance is obtained from the fitting polynomial by  $R_c = 1/q_3$ . In Fig. 6 are shown the I-V curves at different temperatures in  $-0.1$ – $0.1$  V bias range and the fitting curves. The contact resistance is  $5.94 \pm 0.02$  G $\Omega$  at  $55$  °C.

The contact resistances are thermally activated and they are well described by an Arrhenius plot as shown in Fig. 7. From the linear fitting, we obtained for the sample #1 of the detector the value of the barrier height  $\phi_{Bp} = 0.718 \pm 0.004$  eV. For the samples #2 and #3 of the Al/CdTe/Pt detectors, we obtained  $0.733 \pm 0.007$  and  $0.751 \pm 0.008$  eV, respectively.

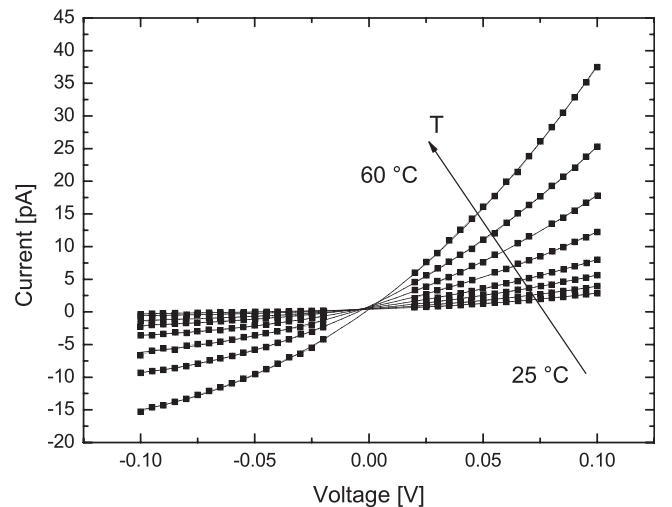


FIG. 6. I-V curves (■) at different temperatures in the  $-0.1$ – $0.1$  V bias range, fitted with a third-degree polynomial (continuous lines).

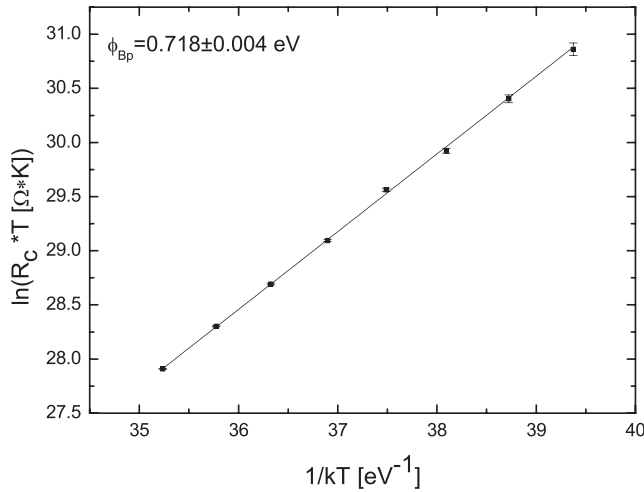


FIG. 7. Arrhenius plot of the contact resistance temperature product for the sample #1 of the Al/p-CdTe/Pt detector.

#### IV. DISCUSSION

The main result of the electrical characterization of the investigated Al/p-CdTe/Pt x-ray detectors is the time dependence of the I-V curves, even for short time intervals ( $<1$  s) between the setting time of the source voltage and the start time of the current measurement. Indeed, the increase with time of current related to polarization effects is a well known critical issue of these Schottky CdTe detectors, contrary to ohmic detectors, but usually this effect on the electrical characterization was taken into account only for long times (i.e.,  $>1$  s).<sup>15,17,18,20,21,31</sup> This time dependence occurs both in forward and reverse bias, except at low voltages ( $<0.1$  V). Generally, the polarization is ascribed to ionization of deep acceptor levels derived from defects (e.g., Cadmium vacancies), which generate a negative space charge near the anode contact after the application of the bias voltage. This space charge screens out the bias and the dead layer is formed near the cathode.<sup>13,14,17,19</sup> Hence, the transport mechanism is dominated in a complicated way by the charge trapping and de-trapping from the deep levels. We noted that at low bias the polarization does not manifest itself and that the dependence of the contact resistance on the temperature obeys Eq. (2) valid for the thermionic emission model. Hence, the extracted barrier height values in the range 0.71–0.73 eV for the investigated samples concerns the barrier for the holes at the Al/CdTe contact. These values are comparable with those obtained by Ref. 26 in Al/p-CdTe diodes, although they obtained two values for the barrier height, 0.69 eV when the thermionic emission model is supposed valid and 0.74 eV with the diffusion model and the capacitance method. Toyama<sup>21</sup> found for Al/p-CdTe/Pt detectors, produced by *Acrorad*, values of the Al/CdTe barrier height in the range 0.2–0.6 eV as a function of the CdTe surface He plasma treatment. In this case, the method to extract the barrier height is based on the assumption that the reverse current, even at high bias voltage, is governed by the thermionic emission model. Moreover, Toyama does not taken into account the time dependence of the I-V curves for short times.

We note that the hole-injecting nature of the CdTe/Pt<sup>15,19,30</sup> contact allow us to neglect its contact resistance and that the carrier transport mechanism is governed both by the Al/CdTe Schottky barrier and by the ionization process of deep levels. Shallow levels do not contribute significantly due to the compensation mechanism.<sup>15,19</sup> The investigated detectors have low values of reverse current, in fact at  $V_R = -1000$  V the pixel current is  $\approx 2$  pA and  $\approx 3$  nA at  $-25^\circ\text{C}$  and  $40^\circ\text{C}$ , respectively. At low temperatures where the effect of polarization is minimized, we note that the reverse current increases with the decrease of the delay time and its value is almost independent of voltage for delay times up to the order of hundred milliseconds.

With delay times  $T_{\text{delay}} \approx 1$  s or greater, where the effect of polarization is present, and for  $V_R < -100$  V and  $T > 0^\circ\text{C}$  the reverse current increases linearly with the voltage, whereas for higher bias voltages, as previously noted, the current shows an exponential trend consistent with the thermionic emission model,<sup>27</sup> although due to the polarization this transport mechanism does not dominate. As a result of the time dependence of the I-V curves shown in Fig. 4 the correct model of the charge transport mechanism in reverse is not clear. In order to clarify this issue we have investigated the time evolution of the reverse current when the detector is switched from zero to  $V_R$  bias. In Fig. 5 are shown the transients at different temperatures for  $V_R = -500$  V within the time window 1 ms–1000 s. After biasing, the current increases so much as to cause the saturation of the amplifier. In this case, the current is dominated by electrons emitted by the cathode.<sup>19</sup> Thereafter, the current decreases due to the emission of trapped charges which causes the reduction of the electric field at the cathode. We note that in Ref. 19 the descending current transient is not observed in the simulated Al/CdTe/Pt Schottky detectors, because fast processes ( $<1$  s) which occur in the detector shortly after biasing are not taken into account.

We observe that the current reaches a minimum value, at the time  $t = t_p$  which we consider the onset of the polarization, thereafter begins to increase with the time. In particular, after the polarization time ( $t > t_p$ ) the current first increases exponentially and then nearly linearly. In this case, the current is governed by the holes sweep out of the depletion volume from the deep acceptors level (detrapping). Hence, from the exponential component of the current we extracted by fitting the detrapping time  $\tau_d$  at each temperature. These  $\tau_d$ 's are thermally activated and they are well described by an Arrhenius plot as shown in Fig. 8, which shows the relation between  $\ln(\tau_d T^2)$  versus  $1/k_B T$ . From the linear fitting of  $\ln(\tau_d T^2)$  versus  $1/k_B T$  we obtained the value of the activation energy  $0.574 \pm 0.001$  eV. This activation energy depends on both the deep acceptor energy levels and the position of the hole quasi-Fermi level which depends on the value of the reverse bias voltage. Indeed, we performed the same analysis at the reverse bias  $V_R = -700$  V, and we obtained an activation energy of  $0.613 \pm 0.001$ . These values of activation energy are comparable to those reported in the previous studies<sup>15,18,22</sup> using different methods in similar CdTe detectors, which are in the range 0.62–0.69 eV. These energy levels are attributed to acceptor levels responsible for

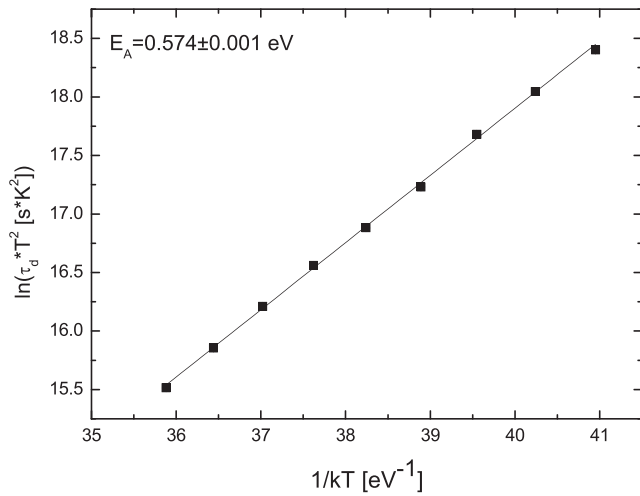


FIG. 8. Arrhenius plot of the time constant  $\tau_d$  obtained from exponential fitting of the reverse current transient at  $V_R = -500$  V and after the polarization time  $t_p$ .

polarization which might depend on the thermal treatment during the crystal growth.<sup>32</sup>

The linear trend of the reverse current for times  $t \gg t_p$  can be explained by considering that the reverse current is dominated by the hole current and is mainly governed by the electrical field at the anode, which increases with the ionization of the deep levels. Thus, the increasing of the reverse current slows the ionization process which causes the slowing down of the speed with which the current increases with the time.<sup>18</sup>

In Fig. 9 is shown the polarization time  $t_p$  versus temperature under bias voltage  $V_R = -500$ , where results the increase of the polarization time ( $t_p$ ) when the detector is cooled. The values of  $t_p$  reported in Ref. 19 for a 1 mm long Al/CdTe/Pt simulated structures, are about two orders of magnitude greater than those shown in Fig. 9. This discrepancy is due to the different definition of the polarization time, indeed in Ref. 19  $t_p$  is defined as the minimum time at which the space charge due to the ionized traps reaches the values imposed by the external bias when the ionization process has not yet started.

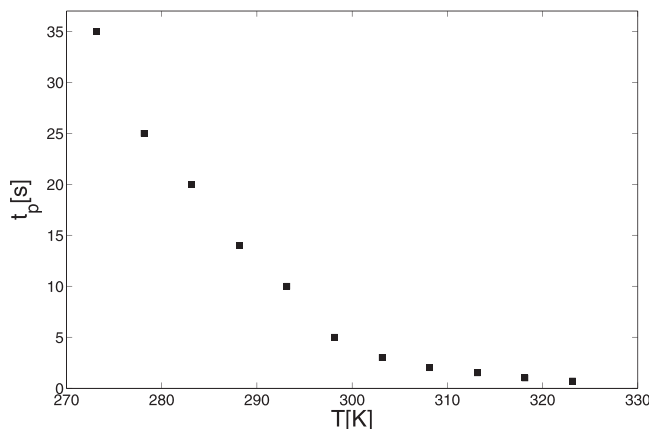


FIG. 9. Temperature dependence of the polarization time  $t_p$  for the Al/p-CdTe/Pt Schottky detector at  $V_R = -500$  V.

Finally, the transient analysis has been also performed at other values of the reverse voltages bias greater than 500 V, and the results showed similar behavior of the transient current.

Concerning the forward conduction mechanism, which is dominated by SCLC, the time dependence of the current-voltage curves complicates the analysis. Indeed, several methods exists (see, for example, Ref. 33) that allow to extract the density of states of the trap levels from the I-V curves analysis, in particular from the dependence of the local  $m$  on the voltage bias  $V_F$ . In our case, the exponent  $m$  depends on the delay time  $T_D$ , hence we leave out the application of these methods.

## V. CONCLUSION

The electrical characterization of Al/p-CdTe/Pt Schottky detectors, fabricated by *Acrorad* for X and gamma ray spectroscopy applications, performed at different temperatures reveals that the current, both in reverse and forward bias, is not only dependent on the temperature but also on the time between the setting time of the source voltage and the starting time of the current measurement. The current-voltage characteristics show the rectifying behavior of these detectors and that the forward current is governed by the time-dependent SCLC mechanism.

The reverse I-V curves exhibit as well time dependence, even at low temperatures. This time dependence of the reverse current occurs in the whole investigated time scale 1 ms–1000 s, where the time evolution of current does not allow to obtain steady-state I-V curves, which generally are used to determine the transport mechanism. This behavior is related to the polarization phenomenon, which is characteristic for Schottky-type CdTe detectors, although usually the emergence of the polarization is assumed to occur a few seconds after the application of the bias voltage.

The time dependence of the I-V disappears for low bias voltage ( $<0.1$  V), where the current is dominated by the thermionic emission model. Indeed, the contact resistance measured at different temperatures is consistent with that predicted by the thermionic emission model and allow us to extract the values of the Al/CdTe Schottky barrier height in the range 0.71–0.73 eV.

## ACKNOWLEDGMENTS

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